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Swallowing accelerometry signal feature variations with sensor displacement

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ABSTRACT

Dual-axis accelerometry has recently shown promise as a non-invasive method for detecting swallowing impairment using signal processing and pattern classification algorithms. However, it is unknown whether variations in sensor placement alter signal characteristics, threatening the accuracy of signal processing classifiers for aspiration detection. To address this question, water swallows were recorded in 14 healthy adults using a dual-axis accelerometer in 13 different positions (baseline, and 2, 4, 6 and 8 mm above, below and to the right of baseline). The baseline position was midline, immediately below the thyroid cartilage during quiet breathing. After segmentation and pre-processing, signal features were extracted in multiple domains (time, frequency, time-frequency). The effect of sensor position on signal feature distributions was examined with non-parametric statistical analysis. The analysis showed that the sensor could be displaced by as much as 4 mm inferior and lateral to the baseline position and by up to 6 mm above the baseline location without significantly altering time-frequency features. In other words, when considering the baseline position as the origin, the admissible region for sensor placement spans 10 mm in the superior-inferior axis and 8 mm in the medial-lateral direction. Results of this study suggest that time-frequency representations of accelerometry signals are most robust to sensor placement variations around the baseline position. The implication of this finding is that a swallowing accelerometry classifier based on time-frequency features can likely tolerate small variations in sensor location without degradation in classification performance.

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1. Introduction

Deglutition or swallowing is the well-defined, complex neuromuscular process for transporting food or liquid from the mouth to the stomach [1,2]. Patients with a range of different neurological disorders such as stroke, cerebral palsy and Parkinson's disease are at high risk of developing swallowing difficulties (dysphagia), where the swallow departs from the well-coordinated typical process [1]. Individuals with dysphagia are susceptible to aspiration, which is defined as the entry of foreign material into the airway below the true vocal folds [2]. If unmanaged, dysphagia can lead to devastating consequences including dehydration, malnutrition, aspiration pneumonia and even death [3,4]. Therefore, identification and careful management of dysphagia are important clinical goals in at-risk populations. Currently the video-fluoroscopic swallowing study (VFSS) is considered to be the gold standard method for dysphagia assessment [2,5]. This method entails the recording of a lateral X-ray video while the patient swallows food or liquid stimuli mixed with barium. The integrity of a swallow is then judged by clinical experts according to criteria such as the depth of airway invasion and the degree of bolus clearance after the swallow. Clinicians then choose appropriate intervention techniques based on their observations of the patients pathophysiology. However, VFSS requires expensive equipment as well as expertise from speech-language pathologists and radiologists [5]. VFSS is also not suitable for day-to-day monitoring due to the exposure to ionizing radiation [6]. Routine monitoring of swallowing function is necessary as the severity of swallowing difficulty can change over time.

Given the above challenges associated with VFSS, several alternative non-invasive techniques to ascertain swallowing function have been investigated, including pulse oximetry [7], cervical auscultation [8], acoustic [9] and electrophysiological methods [10]. An emerging approach is swallowing accelerometry, which entails the attachment of an accelerometer to the neck below the thyroid

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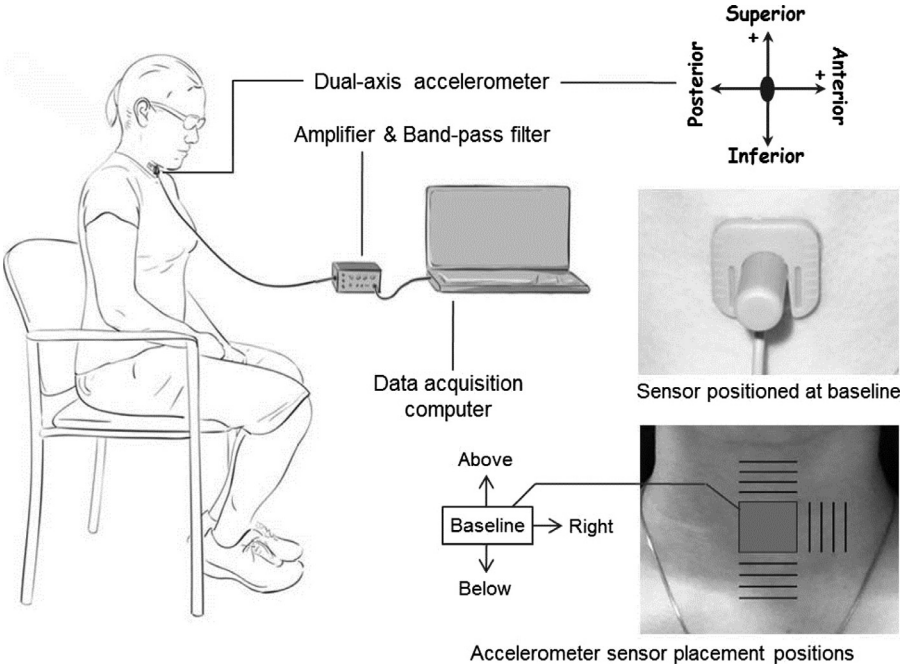


Fig. 1. Experimental setup. The accelerometry sensor, axial orientation, sensor placement positions and signal acquisition instruments are shown.

cartilage to capture the cervical vibrations associated with swallowing. Swallowing accelerometry has been proposed as a method for aspiration detection in several studies [11–16]. Most recently, the potential for automatic aspiration detection based on dual-axis swallowing accelerometry has been demonstrated in a pediatric population with neurological deficits [17] and adult populations with stroke [18–20].

Despite promising findings in the above studies, only signals from the baseline position (midline, immediately below the thyroid cartilage during quiet breathing) have been considered to date. For clinical usability it is important to quantify the impact of slight variations in sensor position on swallowing accelerometry signal features. Therefore, in this study, we compared dual-axis swallowing accelerometry signals recorded in the baseline position with those obtained from twelve different positions emanating from the baseline position. Specifically, we studied swallowing accelerometry signals collected in adults from positions superior to, inferior to and to the right of the baseline position in 2 mm increments. Knowledge of signal feature variations with respect to sensor position will inform protocol designs for clinical swallowing accelerometry measurements as well as the further development of accelerometry-based classifiers of swallowing activity.

2. Experimental paradigm and signal acquisition

2.1. Participants

Fourteen healthy adults (12 females and 2 males) ranging in age from 18–65 years (average 26 ± 8.02 years) with no history of swallowing difficulties, stroke or other neurological conditions, head or neck cancer, spinal cord injury or tracheostomy participated in this study. Participants had a broad distribution of heights (165.51 ± 7.39 cm), weights (63.55 ± 13.24 kg) and neck circumferences (33.32 ± 3.33 cm). The study was approved by the research ethics board of Holland Bloorview Kids Rehabilitation Hospital. Participants provided informed written consent prior to study participation.

2.2. Experimental protocol

Each participant attended two 30-min data collection sessions. In each session, an accelerometer (ADXL322, Analog Devices) was placed on the participant's neck. The axes of the accelerometer were aligned with the superior–inferior and anterior–posterior anatomical axes. In total, thirteen sensor positions were considered. These consisted of the baseline position (midline cervical location with the superior boundary of the sensor aligned with the lower palpable boundary of the thyroid cartilage during quiet breathing) and twelve other positions, i.e., 2, 4, 6 and 8 mm above, below and to the right of the baseline position, as shown in Fig. 1. To keep the protocol within a tolerable time limit, the sensor was only placed at 9 of the 13 different sensor positions (including baseline) for each individual participant. However, across participants, data from all 13 locations were obtained. The skin was marked with non-permanent ink and a flexible measuring tape was used to ensure accurate positioning of the sensor each time. Before sensor attachment at each position, the neck was swabbed with isopropyl alcohol (70%). At each location, the sensor was adhered to the skin using double-sided medical tape.

After attachment of the sensor, the participant was provided with a disposable cup containing 10 ml of water, as measured by a digital scale (Ohaus Adventurer Pro AV812C). Participants were then instructed to empty the contents of the cup into their mouth and swallow, ensuring that their head remained in an upright neutral position while swallowing. Each participant repeated this task 10 times in each sensor position. Participants were asked to initiate the swallowing task upon hearing an audible cue (a verbal instruction to initiate swallow), which was generated 5 s after the beginning of signal recording. Participants were allowed to complete each swallowing task at a comfortable pace.

2.3. Instrumentation

Fig. 1 illustrates the experimental setup, sensor placement positions and axial orientation. In each sensor position, data from the two channels corresponding to the two axes were passed through a pre-amplifier with a bandpass filter (Model P55, Grass Technologies,

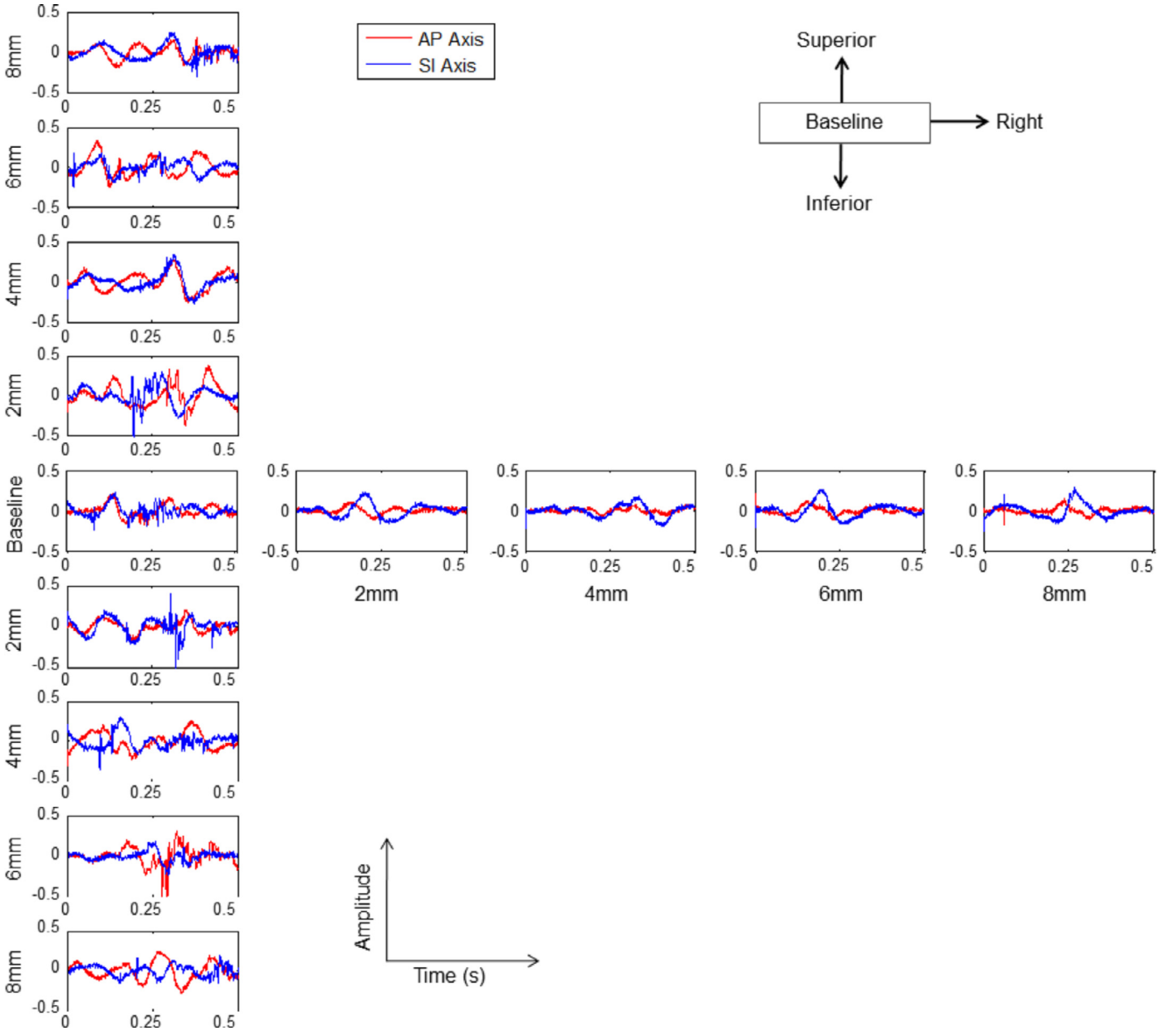


Fig. 2. Examples of segmented anterior-posterior (AP; in red) and superior-inferior (SI; in blue) swallowing accelerometry signals, acquired from baseline and 2 mm, 4 mm, 6 mm and 8 mm superior to, inferior to and to the right of the baseline position. Each graph represents a different swallow. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

West Warwick, RI). The amplifier provided 10-times amplification, and the lower and upper cutoff frequencies of the bandpass filter were set at 0.1 Hz and 3 kHz, respectively. Both signal channels were sampled at 10 kHz through a custom LabVIEW program running on a portable computing device. The use of these filtering, amplification, and sampling specifications has been justified in previous swallow accelerometry studies [18–22]. Acquired data were saved on a computer for further analysis.

3. Methods

3.1. Segmentation and pre-processing

Segmentation and pre-processing of dual-axis accelerometry signals involved extraction of swallowing signal segments and filtering noise from the extracted raw swallow segments, respectively.

Individual swallowing segments were extracted from the continuous swallow sequences using a pseudo-automatic algorithm, which utilizes sequential fuzzy c-means segmentation [23], followed by manual verification. Any swallows that could not be successfully segmented were excluded from the study. In total, 1340 swallow segments were extracted for further analysis. Fig. 2 shows examples of segmented AP and SI swallowing signals acquired from different sensor positions. Note that each graph depicts a different swallow.

Each extracted swallow segment was pre-processed in three steps: (1) whitening, (2) removal of head motion artifacts, and (3) denoising. Whitening was achieved through axis-specific finite impulse response (FIR) filters to annul the unwanted effects introduced by the signal acquisition system (i.e. removal of pink noise) [16]. Next, a spline-based approach was applied to remove the possible effects of head movements as detailed in [24]. Finally, the signals were

Table 1
List of extracted features.

Domain	Axes	
	AP and SI Individually	AP and SI Together
Time	Mean, median, variance, skewness, kurtosis, entropy, regularity, memory, Lempel–Ziv (L–Z) complexity	Mutual information, co-predictability, cross-correlation
Frequency	Peak frequency, central frequency, bandwidth	-
Time-Frequency	Wavelet energy (a10, d10, d9 for AP and SI, d8 and d7 only for SI), wavelet entropy	-

de-noised via a 10-level discrete wavelet decomposition with the discrete Meyer wavelet and soft-thresholding [21,25].

3.2. Feature extraction

To comprehensively estimate the effect of sensor position on swallowing signals, we extracted a wide variety of signal features in time, frequency and time-frequency domains from both AP and SI signals. Within the time and time-frequency domains, a number of information theoretic features were also extracted. In total, 35 features were extracted from each swallowing accelerometry signal. A list of all extracted features in time, frequency and time-frequency domains is found in Table 1. In the following, we detail the computation of less commonly known features and provide literature references as appropriate for the others.

Let $X = \{x_1, x_2, \ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$ be the segmented and pre-processed AP and SI axis accelerometry signals, respectively. For features that were identically extracted from both axes, we only describe the feature derivation from the AP axis signal, X; the same formulation was used to extract features for the SI axis signal, Y. Some features were derived considering both AP and SI signals together.

3.2.1. Time domain features

The sample mean, variance, skewness and kurtosis of the amplitude distribution of each acceleration signal were computed according to the standard definitions of the first four moments of a distribution. Likewise, the median was calculated as a robust estimate of the location of the amplitude distribution.

Entropy. The entropy of an acceleration signal reflects the amount of information or uncertainty in the signal [26]. This measure has been used to describe the behavior of time series in various biomedical signal processing applications [26,27]. The entropy of each standardized acceleration signal (zero mean, unit variance) was calculated according to the formulation described in [14].

Regularity. The extent of regularity in a signal is often quantified as a function of entropy rate [26]. Different from the entropy measure, entropy rate (e.g., approximated entropy or conditional entropy) provides a global indication of the complexity of the distribution. An index of regularity close to unity signifies maximum regularity while an index close to zero indicates maximum complexity or randomness. Regularity was estimated as in [14,20].

Memory. The memory of a signal quantifies the temporal extent of the correlation among nearby samples. To calculate the memory of the acceleration signal, its autocorrelation function was computed from zero to the maximum time lag (equal to the length of the signal) and normalized so that the autocorrelation at zero lag was unity [21]. Memory was estimated as the time interval between zero lag and the point where the autocorrelation becomes less than or equal to $1/e \approx 0.3679$.

L–Z complexity. Lempel–Ziv (L–Z) complexity is a measure of the predictability of a signal [28] and has been applied in the characterization of signal complexity [21,29]. It is a useful scalar estimator of the bandwidth of a random process and the harmonic variability in quasi-periodic signals [29]. L–Z complexity was estimated according as in [20] with 100 levels of quantization.

Mutual information. In dual-axis swallowing accelerometry signals, mutual information reflects the amount of information shared between the AP and SI acceleration signals. If mutual information equals zero, then the vibrations in the AP direction contain no information about the vibrations in the SI direction. Mutual information of the acceleration signal of the AP axis (X) and SI axis (Y) was computed as in [30].

Co-predictability. The predictability of a data point in one signal given a sequence of current and past data points in the other signal can be characterized using cross-entropy rate [31]. Upon estimation of the normalized cross-entropy rate between the AP and SI acceleration signals, an index of co-predictability was derived as in [20]. A co-predictability of 0 signifies that X and Y are completely uncoupled while a co-predictability of 1 indicates that X and Y are perfectly synchronized.

Cross-correlation. The cross-correlation provides a simple measure of dissimilarity between two time series [32]. The cross-correlation between AP (X) and SI (Y) acceleration signals at zero lag, $\hat{R}_{x,y}(0)$, was computed as:

$$\hat{R}_{x,y}(0) = \frac{1}{n}\Sigma_{i=1}^{n} x_i y_i \tag{1}$$

3.2.2. Frequency domain features

To characterize the frequency content of the acceleration signals acquired from different sensor positions, three spectral features were considered. These spectral features are commonly used to analyze audio and biomedical signals and have also been deployed in previous swallowing accelerometry studies [16,21,33].

Peak frequency. The peak magnitude of the fast Fourier transform (FFT) of the acceleration signal, $F_x(f)$, was identified. The corresponding peak frequency, f_p, was computed as:

$$f_p = \underset{f\in[0,f_{\max}]}{\arg\max} |F_X(f)|^2 \tag{2}$$

Centroid frequency. The centroid frequency of the acceleration signal [14] was estimated as follows:

$$\hat{f} = \frac{\int_0^{f_{\max}} f|F_X(f)|^2 df}{\int_0^{f_{\max}} |F_X(F)|^2 df} \tag{3}$$

Bandwidth. The bandwidth of the acceleration signal spectrum was computed as:

$$\mathrm{BW} = \sqrt{\frac{\int_0^{f_{\max}} (f-\hat{f})|F_X(f)|^2 df}{\int_0^{f_{\max}} |F_X(F)|^2 df}} \tag{4}$$

where $F_X(f)$ represents the Fourier transform of the acceleration signal X, f is the centroid frequency, and $f_{\max}$ is the maximum analyzed frequency, i.e. half of the sampling frequency.

Table 2
Features that exhibited significant changes when the sensor was offset (all distances) from the baseline position and (Friedman test, $\alpha = 0.05$). Italicized features exhibited no significant differences in post-hoc pairwise comparisons.

	Direction of offset		
Domain	Superior	Inferior	Right
Time	*Kurtosis AP* Entropy AP Regularity AP Regularity SI	Variance AP Kurtosis AP Variance SI Kurtosis AP Entropy SI Co-predictability AP-SI	Kurtosis AP *Kurtosis SI* Memory SI Regularity SI Cross-correlation AP-SI
Frequency	Centroid frequency SI	*Centroid frequency SI*	Centroid frequency AP *Centroid Frequency SI* *Bandwidth SI*
Time-frequency	*Wavelet energy d_8 AP*	*Wavelet energy d_9 AP* *Wavelet energy a_{10} SI* *Wavelet energy d_7 SI*	Wavelet energy a_{10} SI Wavelet energy d_7 SI Wavelet Entropy SI

3.2.3. Time-frequency domain features

The wavelet transform has the ability to elucidate local, transient, or intermittent information of a signal in both time and frequency planes simultaneously [34]. Thus, both AP and SI axis acceleration signals were decomposed using the discrete Meyer wavelet filter with 10 decomposition levels. The choice of wavelet filter and decomposition levels was selected based on the morphological and spectral properties of the acceleration signals as well as the precedence set by previous dual-axis accelerometry studies [22,25]. From the wavelet decomposition, the following time-frequency features were considered as in [21] and [35].

Wavelet energy. The coefficients of the 10th level wavelet decomposition of an acceleration signal X are denoted as:

$$W_X = [a_{X10}, d_{X10}, d_{X9}, d_{X8}, d_{X7}, d_{X6}, d_{X5}, d_{X4}, d_{X3}, d_{X2}, d_{X1}]$$

where a_{x10} is the vector of approximation coefficients and d_{xl} is the vector of lth-level detail coefficients of the acceleration signal. The wavelet energy in the approximation ($E_{a_{X10}}$) and detail ($E_{d_{Xl}}$) components were defined as follows:

$$E_{ax10} = \frac{||a_{x10}||^2}{E_{T_x}} \times 100\% \quad (5)$$

$$E_{dxl} = \frac{||d_{xl}||^2}{E_{T_x}} \times 100\% \quad (6)$$

$$E_{T_x} = ||a_{X10}||^2 + \Sigma_{l=1}^{10} ||d_{xl}||^2 \quad (7)$$

where $||\cdot||$ represents the Euclidean norm and E_{T_X} is the total energy of the acceleration signal.

The overall wavelet energy in the approximation and detail components was obtained by averaging all swallowing activity acquired from each position across all participants. We observed that decomposition levels corresponding to higher frequencies generally had negligible energy contribution, a finding consistent with previous reports that the majority of swallowing accelerometry signal energy is concentrated in lower frequencies [21,22]. Hence, only the subsets of wavelet energies corresponding to lower frequencies (higher decomposition levels) were retained. Specifically, we accumulated mean energies starting from the lowest spectral band, a_{10}, incrementally adding mean energies from the detail signals until 95% of the total energy was retained. The remaining decomposition levels were excluded from further analysis. As a result, the wavelet energy features corresponding to details, from d_1 to d_8 and from d_1 to d_6 were discarded for the AP and SI axis acceleration signals, respectively.

Wavelet entropy. The wavelet entropy feature was computed by considering the wavelet energy features from all decomposition levels (a_{X10}, $d_{X10,\,9,\,1}$) as follows:

$$WE_x = -\frac{E_{ax10}}{100} \cdot \log_2 \frac{E_{ax10}}{100} - \Sigma_{l=1}^{10} \frac{E_{dxl}}{100} \cdot \log_2 \frac{E_{dxl}}{100} \quad (8)$$

This feature characterizes the underlying complex functional dynamics of the accelerometry signal in both time and frequency domain [35]. In contrast to temporal or spectral entropy measure alone, wavelet entropy quantitatively evaluates the temporally localized information content of the multi-frequency non-stationary signals [35].

4. Statistical analysis

We combined data from all participants to identify the effects of sensor position on each extracted feature. For each feature, baseline feature values were compared against those superior to, inferior to and to the right of baseline, at all four distances (2 mm, 4 mm, 6 mm and 8 mm). Due to the non-normality of the feature value distributions (as per the Lilliefors test for normality), the nonparametric Friedman test was employed [35,36] with $\alpha = 0.05$. Post-hoc analyses for features with significant position effects were performed using Wilcoxon signed rank tests with $\alpha = 0.05$ [36,37]. No correction was applied to the significance level so as to render the test more sensitive to differences between offset and baseline sensor positions.

5. Results

Table 2 depicts the features with significant position effects in directions superior to, inferior to and to the right of baseline, respectively, according to the Friedman tests. In the superior direction, only six of the 35 extracted features for AP and SI acceleration signals exhibited significant changes from baseline values. On the other hand, only 10 and 11 of the 35 features changed significantly from baseline when the sensor was positioned inferior to and to the right of baseline, respectively. In other words, the majority of the features considered in the present study were not significantly affected by deviations in sensor position. Time-frequency domain features were generally more robust to positional offsets than time and frequency domain features.

In the post-hoc analysis, pairwise feature comparisons between baseline and each of the offset positions (2, 4, 6 and 8 mm) in each direction indicated that some features (two in the superior, four in the inferior and three in the right direction, marked in italics in Table 2) were robust to positional offsets. Others, however, exhibited significant differences between baseline and each offset positions as

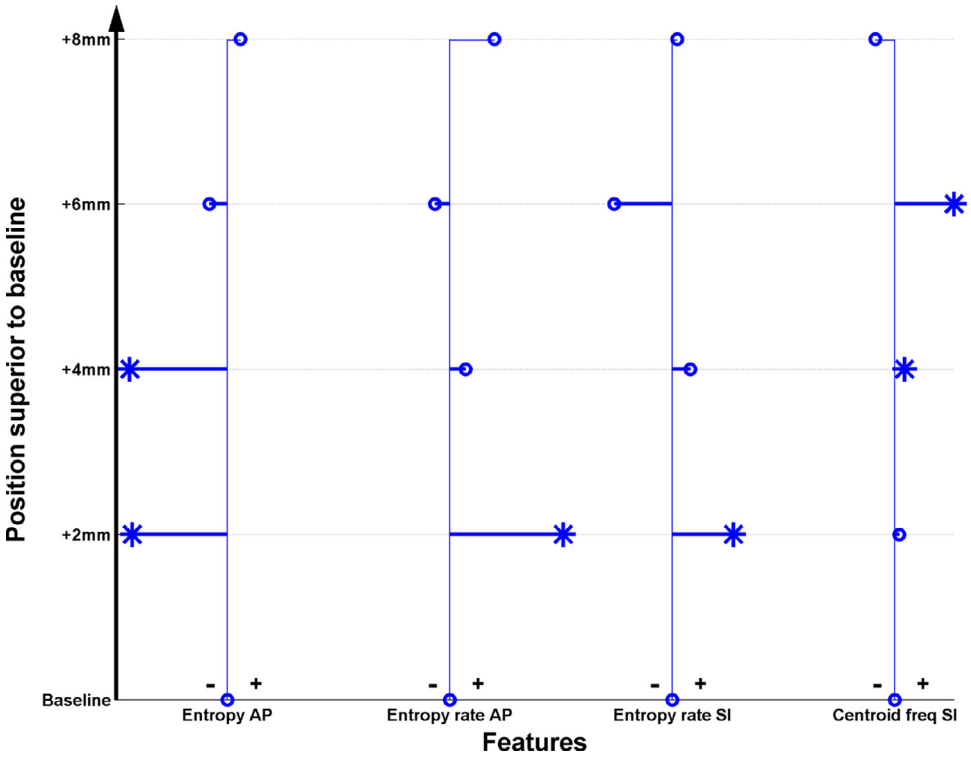


Fig. 3. Stem plot denoting differences between median feature values at baseline and superior offset positions (Wilcoxon signed rank test with $\alpha = 0.05$). Significant differences from baseline values are indicated by an asterisk. Stems to the left or right of each feature axis denote decrease ($-$) or increase (+), respectively, from the baseline median.

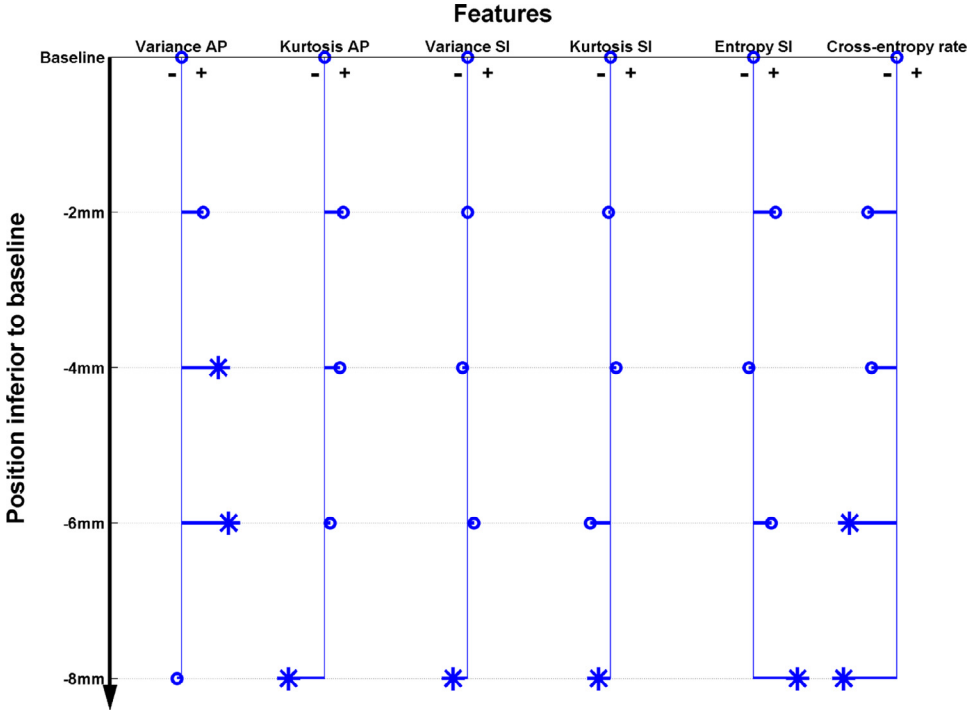


Fig. 4. Stem plot denoting differences between median feature values at baseline and inferior offset positions (Wilcoxon signed rank test with $\alpha = 0.05$). Significant differences from baseline values are indicated by an asterisk. Stems to the left or right of each feature axis denote decrease ($-$) or increase (+), respectively, from the baseline median.

depicted in the stem plots of Figs. 3–5. In each of these figures, one axis represents the positional offset and the other catalogs the features whose median departs from baseline values in at least one position. Each stem denotes the difference between the median feature values at baseline and at an offset position, normalized by the geometric mean of the corresponding interquartile ranges. The lines parallel to the positional axis thus signify 0 difference between baseline and offset values. Both negative (median at offset position $<$ median at baseline) and positive (median at the offset position $>$ baseline median) are shown.

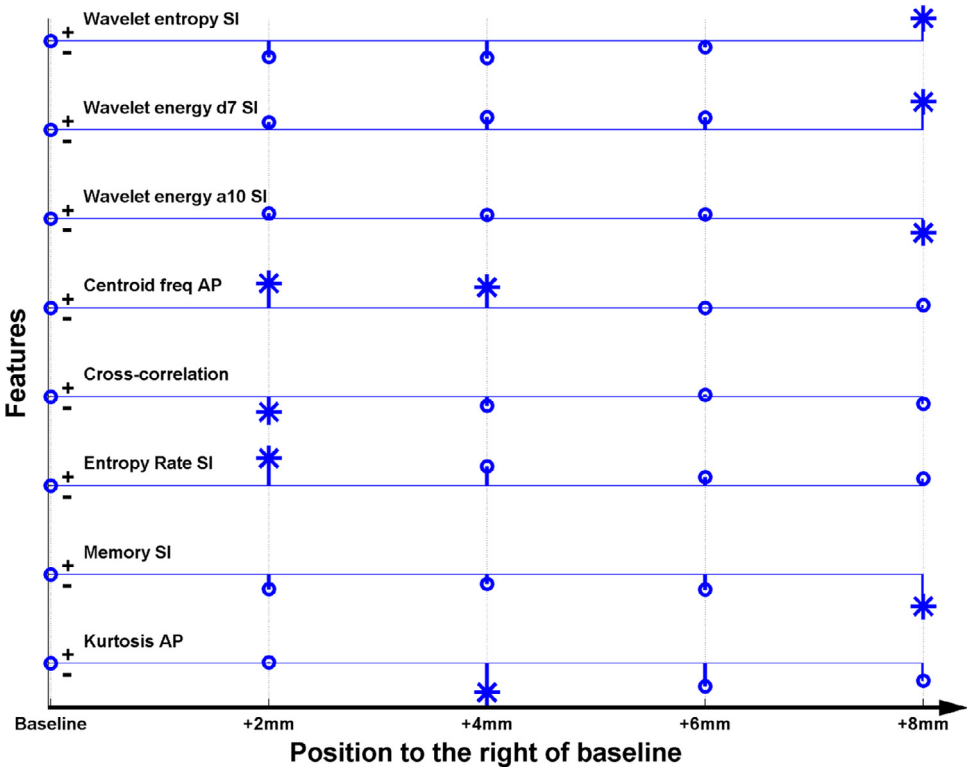


Fig. 5. Stem plot denoting differences between median feature values at baseline and laterally offset positions (Wilcoxon signed rank test with $\alpha = 0.05$). Significant differences from baseline values are indicated by an asterisk. Stems below and above each feature axis denote decrease (−) or increase (+), respectively, from the baseline median.

6. Discussion

6.1. Time domain features

From Table 2 and Figs. 3–5, it is evident that time domain features were the most susceptible to change with sensor positioning. In particular, entropy-related information theoretic features varied in all three directions of offset. Interestingly, entropy decreased (in the AP direction) while regularity increased in both AP and SI directions, suggesting that the signal became less random and more regular for small superior offsets but not for greater offsets. This observation may be explained by consulting morphometric studies of laryngeal cartilage and reflecting upon the biomechanical properties of the underlying soft tissues. In the baseline position, the sensor was most likely situated over the inferior aspect of the thyroid cartilage and partially over the medial cricothyroid ligament. Please refer to Fig. 6 for an anatomical reference. Given that the anterior thyroid cartilage height is reportedly 16.4 ± 2.7 mm in males and 13.4 ± 3.2 mm in females [38], the positions superior to baseline likely approached the laryngeal prominence. The corresponding bias towards larger signal amplitudes (as suggested by the significant increase in AP signal variance) may explain the observed decrease in AP entropy (signal becomes less uniform random) and an increase in AP and SI entropy rates. This escalating variance would also explain the lack of statistical differences in temporal features between baseline and the most superior offsets.

In the most inferior position, the sensor was likely over inferior edge of the cricoid cartilage and perhaps bordering tracheal cartilage. Note that the thyroid cartilage is anatomically closer to the hyoid bone and hyolaryngeal excursion has been implicated as the primary physiological source of swallowing accelereometry signals [39]. This finding, combined with the fact that the thyroid cartilage likely has a higher moment of inertia by virtue of its greater mass, may explain the observed significant decrease in SI signal variance in the position

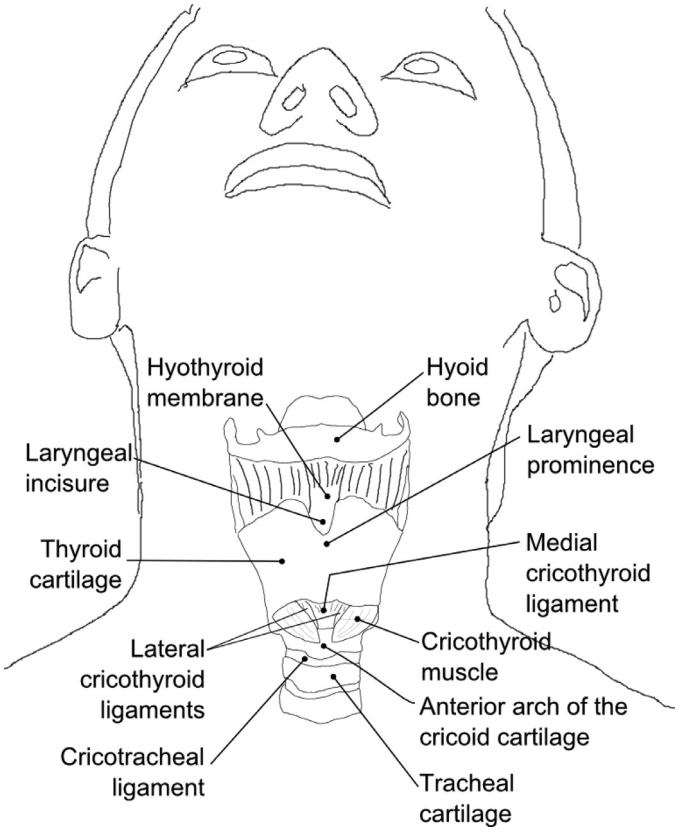


Fig. 6. Laryngeal anatomy. In the baseline position, the sensor was situated over the inferior aspect of the thyroid cartilage and partially over the medial cricothyroid ligament.

most inferior to baseline. SI kurtosis also decreased, implying a more platykurtic amplitude distribution. This would in turn account for the observed increase in SI entropy (i.e., more uniform amplitude distribution).

The cross-entropy rate decreases below baseline values at 6 and 8 mm offsets, meaning that the AP and SI signals became more decoupled. This decoupling may be attributed in part, to the reduced upward motion of cricoid/tracheal cartilage, as suggested by the significant decrease in SI signal variance at the most extreme inferior offset. Furthermore, the increased SI entropy suggests that the SI signal became more random, supporting the notion that AP and SI signals became decoupled at the 8 mm inferior offset.

In terms of lateral offsets, SI memory decreased in the most lateral position, where the sensor likely rested over a greater extent of thyroid cartilage than at baseline. This finding can be explained by the fact that cartilage is less elastic than ligament [40], and thus instantaneous vibrations would have lower correlation with those at successive times, leading to shorter memory.

6.2. Frequency domain features

In the superior direction, only the centroid frequencies of the SI axis in the 4 and 6 mm offset positions were significantly higher than that of baseline. This could be explained by the fact at these locations, the sensor was entirely situated over the thyroid cartilage, which being a stiffer material than ligament, could vibrate at higher frequencies. By the same token, as the sensor continued to travel upwards, it may have reached the laryngeal incisure or even the median hyothyroid ligament, in which case higher frequencies would be damped by ligament. This would explain the lack of difference between the most superior and baseline positions. The only other frequency domain feature to vary from baseline was the centroid frequency of the AP axis signal, which increased above the baseline value at the 2 and 4 mm lateral offset positions. At these positions, the sensor may have veered off the cricothyroid ligament and partially onto the circothyroid muscle, in which case, the viscoelasticity and hysteresis of ligament [41] may have had less of a dampening effect on AP vibrations. At the most lateral offsets, the sensor could be situated over the intersection of the cricothyroid muscle and cricoid cartilage, leading to the observed increase in the coefficient of variation of the centroid frequency and thus a lack of difference from baseline.

6.3. Time-frequency features

The wavelet energy at the 10th level approximation (a10) decreased significantly in the most lateral position. This decrease in low frequency energy was accompanied by a corresponding increase in the wavelet energy at the 7th level detail (d7). Thus, there seems to be a shift in signal energy from low to high frequencies at the most lateral offset. As suggested above, in the most lateral position, the sensor probably rested primarily over the thyroid cartilage, and minimally over ligament (in contrast to the baseline position). The elasticity differences of cartilage and ligament would explain the observed spectral shift to higher frequencies.

Finally, at the 8 mm lateral offset, SI wavelet entropy significantly increased beyond basal values. This can be attributed to the corresponding decrease in approximation energy and increase in detail energy, both of which contribute to an increase in wavelet entropy. This counterintuitive result is due to the fact that the constituent function $-x \log x$ for wavelet entropy has an inverse parabolic shape for $0 \leq x \leq 1$, attaining a maximum value at $x = 10^{-\frac{1}{\ln 10}} \approx 0.37$ Furthermore, the approximation energy typically dominates the signal energy (i.e., greater than 37%) and thus a decrease in approximation energy increases wavelet entropy. The converse applies for the detail signals.

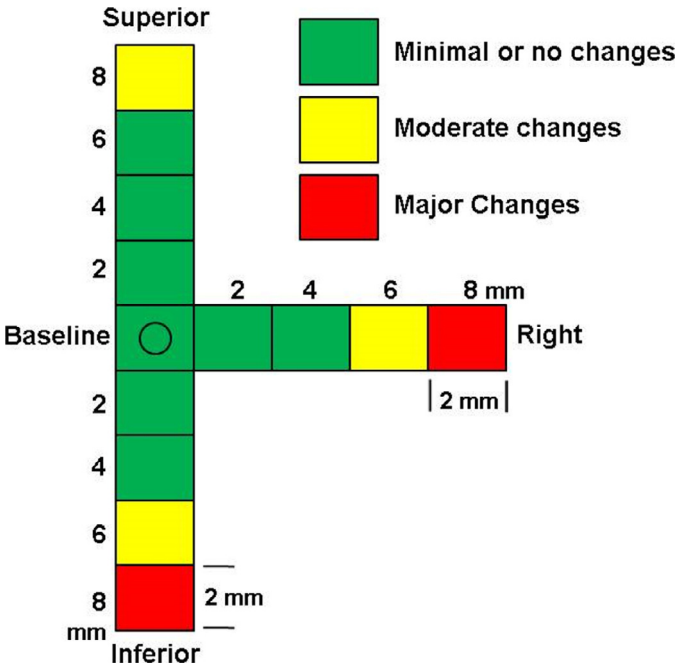


Fig. 7. Accelerometry sensor positions showing significant feature variations (green: minimal or no changes, yellow: moderate changes and red: major changes) with the displacement of sensor positions in superior, inferior and right directions from baseline. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 7 presents a topographic summary of our findings. Collectively, our analyses indicate that the accelerometry sensor could be displaced by up to 4 mm below or to the right of the baseline position or by up to 6 mm above baseline without significantly altering the signal features studied herein, all of which have been deployed in previous swallowing studies. Accordingly, we can conclude that accelerometry signals collected from the front of the neck, just below the thyroid cartilage, are similar to those collected in a region around this position, spanning approximately 10 mm along the SI axis and 8 mm in the medial-lateral direction, assuming anatomical symmetry. Our findings support the use of a recently developed classification algorithm that relied exclusively on time-frequency features for identifying aspiration [18].

It is important to note that all the above observations pertain only to thin consistency boluses. Further research is needed to quantify the effects of sensor displacement on swallowing accelerometry related to higher viscosity boluses. Moreover, our study only pertains to swallowing in young adults without swallowing difficulties. Positional offsets may have different effects when dealing with loose skin or excessive adipose tissue, as may be present in other populations.

7. Conclusions

We critically examined variations in swallowing accelerometry signals acquired from various locations around a baseline position. Our findings suggest that swallowing signal features for thin consistency boluses are generally invariant to small positional offsets (10 mm in the superior-inferior direction and 8 mm in the medial-lateral direction, assuming anatomical symmetry) around the baseline position. Time-frequency domain features, in particular, are the most robust to variations of sensor placement.

Ethical approval

Ethical approval was provided by the Research Ethics Board of Holland Bloorview Kids Rehabilitation Hospital, Toronto, Canada. Research Ethics Board Protocol #12-315.

Conflict of interest

None declared.

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